

Group I, Row V, part 2b: the UV terms and the other terms of $\frac{d^3 N_{2b}^{\text{UV}}}{d^3 k}$

Starting from your expression, with nothing dropped: the pole in $d = 2 - 2\epsilon$, the finite remainder in closed form, and the p^+ integral with the plus prescription — each step checked numerically

$$r = z - x, \quad s = y - x, \quad P = x' - w', \quad \xi = \frac{p^+}{k^+}, \quad \bar{\xi} = 1 - \xi, \quad q = \bar{\xi}k, \quad M^2 = \frac{\xi}{\bar{\xi}} s^2, \quad D = (r - s)^2 + M^2, \quad D_0 = D|_{r=0} = \frac{s^2}{\bar{\xi}}, \quad \lambda = \frac{\Lambda}{k^+}, \quad L = \log \frac{k^+}{\Lambda},$$

$$C_{\text{UV}}(\xi) = \xi \bar{\xi} + \frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi}, \quad \mathcal{W} = \frac{P \cdot s}{P^2 s^2}, \quad \ell = \log \frac{k^2 s^2 e^{2\gamma_E}}{4}$$

The answer. With $\mathcal{N} = \frac{1}{(2\pi)^3} \frac{g^4}{16\pi^4} \frac{1}{k^+}$ and $\mathcal{U}_b = [U^{ab'}(x') - U^{ab'}(w')] [-N_c U^{af}(x) + N_c U^{af}(y)] \rho^{b'}(x') \rho^f(y)$:

$$\frac{d^3 N_{2b}^{\text{UV}}}{d^3 k} = \mathcal{N} \int_{x,y,x',w'} e^{-ik \cdot w'} e^{ik \cdot x} \mathcal{W} \left\{ \underbrace{\left(2L - \frac{11}{6}\right) \left[\frac{1}{\epsilon_{\text{UV}}} + \log \frac{\mu_{\text{MS}}^2 (y-x)^2 e^{2\gamma_E}}{4} \right]}_{\text{UV term}} + \underbrace{\frac{1 + e^{ik \cdot (y-x)}}{2} L^2 + c_0 L + H}_{\text{other terms}} \right\} \mathcal{U}_b$$

- There is exactly one UV pole, with coefficient $2L - \frac{11}{6}$.
- c_0 (eq. 6.4) and H (eq. 7.3) are finite, dimensionless numbers that depend only on $k \cdot (y - x)$ and $|k||y - x|$. Each is given as an explicit one-dimensional integral.
- **Warning.** The $\frac{1}{2} e^{ik \cdot (y-x)} L^2$ is an artefact of this split. The rest of 2b carries exactly $-\frac{1}{2} e^{ik \cdot (y-x)} \mathcal{W} L^2$, so the two cancel (§9). The whole of 2b keeps only $\frac{1}{2} \mathcal{W} L^2$.

0. Your expression, checked, and rewritten

Prefactor. $\frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{1}{k^+} \cdot \frac{1}{2\pi} = \frac{1}{(2\pi)^3} \frac{g^4}{16\pi^5} \frac{1}{k^+} = \frac{\mathcal{N}}{\pi}$. This agrees with the .tex start of part (b).

Colour. $[-N_c U^{af}(x) + N_c U^{af}(y)] = N_c [U^{af}(y) - U^{af}(x)]$, so the colour factor is $\mathcal{U}_b$. It has no z and no ξ dependence.

Phase. $e^{-ik \cdot w'} e^{ik \cdot x} e^{iq \cdot r}$ is correct, with $q = \bar{\xi}k$.

D . Your $D_0 = s^2 + M^2 = \frac{s^2}{\bar{\xi}}$ is the value of $D = (r - s)^2 + M^2$ at $r = 0$, with $M^2 = \frac{\xi}{\bar{\xi}} s^2$. Check: $s^2 + \frac{\xi}{\bar{\xi}} s^2 = \frac{\xi + \bar{\xi}}{\bar{\xi}} s^2 = \frac{s^2}{\bar{\xi}}$.

Rewrite. Since $\frac{s \cdot P}{P^2 r^2} = \mathcal{W} \frac{s^2}{r^2}$, all the P dependence sits in $\mathcal{W}$, and

$$\frac{d^3 N_{2b}^{\text{UV}}}{d^3 k} = \mathcal{N} \int_{x,y,x',w'} e^{-ik \cdot w'} e^{ik \cdot x} \mathcal{W} \int_{\lambda}^{1-\lambda} d\xi C_{\text{UV}}(\xi) \frac{1}{\pi} K(\xi) \mathcal{U}_b, \quad K(\xi) \equiv \mu^{-2\epsilon} \int d^d r e^{iq \cdot r} \frac{s^2}{\xi D r^2}.$$

K depends only on ξ , k and s . The whole problem is this one scalar integral.

1. Does this piece contain all the UV divergence of 2b?

Yes. As $r \rightarrow 0$ the full 2b integrand goes to $\frac{T_{\text{UV}}}{s^2}$, where

$$T_{\text{UV}} = d_{\perp} \xi \bar{\xi} \frac{(P \cdot r)(s \cdot r)}{P^2 r^4} + \left(\frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} \right) \frac{P \cdot s}{P^2 r^2}.$$

Your $\xi \bar{\xi}$ term is $\xi \bar{\xi} \frac{P \cdot s}{P^2 r^2}$ instead of the tensor. The pole comes from the angular average, and $\langle \hat{r}^i \hat{r}^m \rangle = \frac{\delta^{im}}{d}$. So $d_{\perp} \langle (P \cdot \hat{r})(s \cdot \hat{r}) \rangle = \frac{d}{d} P \cdot s = P \cdot s$, since $d_{\perp} = d$. The two therefore have the same pole.

The difference, $\xi \bar{\xi} \frac{2(P \cdot r)(s \cdot r) - (P \cdot s)r^2}{P^2 r^4} \cdot \frac{e^{iq \cdot r}}{\xi D}$, is traceless and UV finite. It belongs to the *rest* of 2b, where it can be computed at $d_{\perp} = d = 2$. **Your split is valid as long as the rest of 2b is defined as (full 2b) – (this piece).**

2. The regions of the r integral

region	behaviour of $\frac{s^2}{\xi D r^2}$	verdict
$r \rightarrow 0$	$D \rightarrow D_0 = \frac{s^2}{\xi}$, so $\frac{s^2}{\xi D} \rightarrow 1$ and the integrand $\rightarrow \frac{1}{r^2}$; $\int d^2 r \frac{1}{r^2}$ diverges logarithmically	UV divergent
$r \rightarrow s$	$\frac{1}{D} = \frac{1}{(r-s)^2 + M^2}$ with $M > 0$	finite, but $\sim \log \frac{1}{M^2}$ as $\xi \rightarrow 0$ (§6)
$ r \rightarrow \infty$	$\frac{s^2}{\xi r^4}$	finite, even without the phase

3. Exact decomposition of K

3.1 An algebraic identity

Expand $D = r^2 - 2r \cdot s + s^2 + M^2 = r^2 - 2r \cdot s + D_0$. Hence $D_0 = D + 2r \cdot s - r^2$, and with $\bar{\xi} D_0 = s^2$:

$$\frac{s^2}{\bar{\xi} D r^2} = \frac{D_0}{D r^2} = \frac{D + 2r \cdot s - r^2}{D r^2} = \frac{1}{r^2} + \frac{2r \cdot s}{r^2 D} - \frac{1}{D}.$$

So

$$K(\xi) = \underbrace{\mu^{-2\epsilon} \int d^d r \frac{e^{iq \cdot r}}{r^2}}_{M_0 \text{ (pole)}} + \underbrace{\int d^2 r e^{iq \cdot r} \frac{2r \cdot s}{r^2 D}}_{A \text{ (finite)}} - \underbrace{\int d^2 r \frac{e^{iq \cdot r}}{D}}_{B \text{ (finite)}}.$$

A and B are finite:

- A : $\frac{r \cdot s}{r^2} \sim \frac{1}{r}$ at small r , and $\sim \frac{1}{r^3}$ at large r ;
- B : $D \geq M^2$ and the phase controls the $\frac{1}{r^2}$ tail.

So they are computed at $d = 2$. All of the ϵ dependence is in M_0 .

3.2 The pole term M_0

From $\int d^d r e^{iq \cdot r} (r^2)^{-1} = \pi \Gamma(-\epsilon) \left(\frac{q^2}{4\pi}\right)^\epsilon$ and $\mu_{\overline{\text{MS}}}^2 = 4\pi\mu^2 e^{-\gamma_E}$:

$$M_0 = \pi \left[\frac{1}{\epsilon_{\text{UV}}} + \log \frac{\mu_{\overline{\text{MS}}}^2}{q^2} \right], \quad \frac{1}{\epsilon_{\text{UV}}} \equiv -\frac{1}{\epsilon}.$$

In transverse coordinates (rowV_coordinate_UV.pdf, §4 and (R1)), write $M_0 = s^2 J_0 + \int d^2 r \frac{1}{r^2} [e^{iq \cdot r} - \frac{s^2}{r^2 + s^2}]$. This uses

$$s^2 J_0 = \pi \left[\frac{1}{\epsilon_{\text{UV}}} + \log \frac{\mu_{\overline{\text{MS}}}^2 s^2 e^{2\gamma_E}}{4} \right], \quad \int d^2 r \frac{1}{r^2} \left[e^{iq \cdot r} - \frac{s^2}{r^2 + s^2} \right] = -\pi \log \frac{q^2 s^2 e^{2\gamma_E}}{4}.$$

The two add up to the same M_0 . The second line is $2\pi \int_0^\infty \frac{dx}{x} [J_0(q|x|x) - \frac{1}{1+x^2}]$.

3.3 The term B

Shift $r = s + u$, so that $D = u^2 + M^2$ and $e^{iq \cdot r} = e^{iq \cdot s} e^{iq \cdot u}$. The standard 2D Fourier transform $\int d^2 u \frac{e^{iq \cdot u}}{u^2 + M^2} = 2\pi \int_0^\infty \frac{u du J_0(qu)}{u^2 + M^2} = 2\pi K_0(qM)$ gives

$$B = 2\pi e^{iq \cdot s} K_0(qM), \quad qM = \bar{\xi} |k| \cdot |s| \sqrt{\frac{\xi}{\bar{\xi}}} = |k| |s| \sqrt{\xi \bar{\xi}}.$$

3.4 The term A (Feynman parameter)

Combine the denominators. Use $\frac{1}{ab} = \int_0^1 \frac{dt}{[ta + (1-t)b]^2}$ with $a = D$ and $b = r^2$:

$$tD + (1-t)r^2 = r^2 - 2tr \cdot s + tD_0 = (r - ts)^2 + \Delta, \quad \Delta = tD_0 - t^2 s^2 = t s^2 \left(\frac{1}{\xi} - t \right) = t s^2 \left(1 - t + \frac{\xi}{\bar{\xi}} \right) > 0.$$

Shift $r = ts + u$. Then $2r \cdot s = 2u \cdot s + 2ts^2$ and $e^{iq \cdot r} = e^{itq \cdot s} e^{iq \cdot u}$.

$$A = \int_0^1 dt e^{itq \cdot s} \int d^2 u e^{iq \cdot u} \frac{2u \cdot s + 2ts^2}{(u^2 + \Delta)^2}.$$

Two Fourier transforms, both obtained from $\int d^2 u \frac{e^{iq \cdot u}}{u^2 + \Delta} = 2\pi K_0(q\sqrt{\Delta})$:

- Differentiate in Δ , using $K'_0 = -K_1$:

$$\int d^2 u \frac{e^{iq \cdot u}}{(u^2 + \Delta)^2} = -\partial_\Delta [2\pi K_0(q\sqrt{\Delta})] = \frac{\pi q}{\sqrt{\Delta}} K_1(q\sqrt{\Delta}).$$

- Write $\frac{u^j}{(u^2 + \Delta)^2} = -\frac{1}{2} \partial_j \frac{1}{u^2 + \Delta}$ and integrate by parts ($\int e^{iq \cdot u} \partial_j f = -iq^j \int e^{iq \cdot u} f$):

$$\int d^2 u e^{iq \cdot u} \frac{u^j}{(u^2 + \Delta)^2} = \frac{iq^j}{2} 2\pi K_0(q\sqrt{\Delta}) = i\pi q^j K_0(q\sqrt{\Delta}).$$

Therefore

$$A = \pi \int_0^1 dt e^{itq \cdot s} \left[2i (q \cdot s) K_0(q\sqrt{\Delta}) + \frac{2ts^2 q}{\sqrt{\Delta}} K_1(q\sqrt{\Delta}) \right].$$

3.5 Check against direct integration

In $d = 2$, $\int d^2 r \frac{1}{r^2} [e^{iq \cdot r} \frac{s^2}{\xi D} - \frac{s^2}{r^2 + s^2}]$ is absolutely convergent. By 3.1–3.4 it must equal $-\pi \log \frac{q^2 s^2 e^{2\gamma_E}}{4} + A - B$. Test case: $s = (0.7, -0.4)$, $k = (1.3, 0.5)$:

ξ	direct 2D quadrature	closed form
0.3	2.52869189 + 2.18595267 i	2.52869143 + 2.18595268 i
0.6	3.12302159 + 0.97391230 i	3.12302120 + 0.97391230 i

4. The UV term and the other terms, separated

Put 3.2 (coordinate form), 3.3 and 3.4 into $K = M_0 + A - B$ and divide by π :

$$\begin{aligned} \frac{K(\xi)}{\pi} &= \underbrace{\frac{1}{\epsilon_{UV}} + \log \frac{\mu_{\overline{\text{MS}}}^2 (y-x)^2 e^{2\gamma_E}}{4}}_{\text{UV term}} + \underbrace{G(\xi)}_{\text{other terms}}, \\ G(\xi) &= -\log \frac{\bar{\xi}^2 k^2 s^2 e^{2\gamma_E}}{4} + a(\xi) - 2 e^{i\bar{\xi} k \cdot s} K_0(|k||s| \sqrt{\xi \bar{\xi}}), \\ a(\xi) &= \frac{A}{\pi} = \int_0^1 dt e^{it\bar{\xi} k \cdot s} \left[2i\bar{\xi} (k \cdot s) K_0(\bar{\xi}|k|\sqrt{\Delta}) + \frac{2t\bar{\xi}|k|s^2}{\sqrt{\Delta}} K_1(\bar{\xi}|k|\sqrt{\Delta}) \right], \quad \Delta = ts^2 \left(\frac{1}{\bar{\xi}} - t \right). \end{aligned}$$

(If you prefer the momentum form, use $\log \frac{\mu^2 (y-x)^2 e^{2\gamma_E}}{4} = \log \frac{\mu^2}{k^2} + \ell$ and move ℓ into the other terms.) So, before the p^+ integral,

$$\frac{d^3 N_{2b}^{\text{UV}}}{d^3 k} = \mathcal{N} \int e^{-ik \cdot w'} e^{ik \cdot x} \mathcal{W} \int_\lambda^{1-\lambda} d\xi C_{UV}(\xi) \left\{ \frac{1}{\epsilon_{UV}} + \log \frac{\mu_{\overline{\text{MS}}}^2 (y-x)^2 e^{2\gamma_E}}{4} + G(\xi) \right\} \mathcal{U}_b.$$

5. p^+ integral of the UV term

The UV bracket does not depend on ξ . So we only need

$$\int_\lambda^{1-\lambda} d\xi C_{UV} = \frac{1}{6} + (L-1) + (L-1) = 2L - \frac{11}{6},$$

using $\int_\lambda^{1-\lambda} \frac{\xi}{\xi} = \int_\lambda^{1-\lambda} \left(\frac{1}{\xi} - 1 \right) = \log \frac{1-\lambda}{\lambda} - (1-2\lambda) \rightarrow L-1$, and the same for $\frac{\bar{\xi}}{\xi}$.

6. The endpoints of $G(\xi)$

The plus prescription needs to know how $C_{UV}G$ behaves at each end. We have $C_{UV} \rightarrow \frac{1}{\xi}$ at $\xi \rightarrow 0$ and $C_{UV} \rightarrow \frac{1}{\xi}$ at $\xi \rightarrow 1$. Throughout we use $K_0(z) = -\log \frac{ze^{\gamma_E}}{2} + O(z^2 \log z)$ and $zK_1(z) = 1 + O(z^2 \log z)$.

6.1 $\xi \rightarrow 1$ ($\bar{\xi} \rightarrow 0$)

- $-\log \frac{\bar{\xi}^2 k^2 s^2 e^{2\gamma_E}}{4} = 2 \log \frac{1}{\bar{\xi}} - \ell$.
- $qM = |k||s|\sqrt{\xi\bar{\xi}} \rightarrow 0$, so $2K_0(qM) = -\log \frac{k^2 s^2 \xi \bar{\xi} e^{2\gamma_E}}{4} = \log \frac{1}{\bar{\xi}} - \ell + O(\bar{\xi})$, and $e^{i\bar{\xi}k \cdot s} \rightarrow 1$.
- $a \rightarrow 0$. Since $zK_1(z) \leq 1$, the K_1 term is bounded by $\frac{2ts^2}{\Delta} = \frac{2}{1/\xi - t} \leq \frac{2\xi}{\xi} \rightarrow 0$, and the K_0 term is $O(\bar{\xi} \log \bar{\xi})$.

$$\Rightarrow G \xrightarrow{\xi \rightarrow 1} 2 \log \frac{1}{\bar{\xi}} - \ell - \left(\log \frac{1}{\bar{\xi}} - \ell \right) = \log \frac{1}{\bar{\xi}} \quad (\text{constant term exactly } 0). \quad (6.1)$$

Meaning. For $\bar{\xi} \rightarrow 0$ the integrand stays $\approx \frac{1}{r^2}$ from $|s|$ out to $M = \frac{|s|}{\sqrt{\bar{\xi}}}$, which gives $\int_{|s|}^M \frac{2\pi r dr}{r^2} = \pi \log \frac{1}{\bar{\xi}}$.

$\bar{\xi}$	$G(\xi)$	$\log \frac{1}{\bar{\xi}}$	difference
10^{-2}	$4.59435 + 0.00707i$	4.60517	$1.3 \cdot 10^{-2}$
10^{-4}	$9.21009 + 0.00007i$	9.21034	$2.6 \cdot 10^{-4}$
10^{-6}	$13.815507 + 0.000001i$	13.815511	$4.1 \cdot 10^{-6}$

6.2 $\xi \rightarrow 0$

- $q \rightarrow k$, so $-\log \frac{q^2 s^2 e^{2\gamma_E}}{4} \rightarrow -\ell$.
- $M^2 \rightarrow \xi s^2$, so $2K_0(qM) \rightarrow \log \frac{1}{\xi} - \ell$ and $e^{iq \cdot s} \rightarrow e^{ik \cdot s}$.
- In $a(\xi)$, $\Delta \rightarrow ts^2(1 - t + \xi)$ becomes small near $t = 1$. There the K_1 term $\rightarrow \frac{2ts^2}{\Delta} \rightarrow \frac{2}{1-t+\xi}$, and $\int_0^1 \frac{2dt}{1-t+\xi} = 2 \log \frac{1+\xi}{\xi}$. Subtracting this and adding it back:

$$a(\xi) \xrightarrow{\xi \rightarrow 0} 2e^{ik \cdot s} \log \frac{1}{\xi} + a_{\text{reg}}, \quad a_{\text{reg}} = \int_0^1 dt \left\{ e^{itk \cdot s} \left[2i(k \cdot s)K_0(|k|\sqrt{\Delta_0}) + \frac{2t|k|s^2}{\sqrt{\Delta_0}} K_1(|k|\sqrt{\Delta_0}) \right] - \frac{2e^{ik \cdot s}}{1-t} \right\},$$

with $\Delta_0 = t(1-t)s^2$. The braces are integrable at $t = 1$. Adding up:

$$G \xrightarrow{\xi \rightarrow 0} -\ell + 2e^{ik \cdot s} \log \frac{1}{\xi} + a_{\text{reg}} - e^{ik \cdot s} \left(\log \frac{1}{\xi} - \ell \right) = e^{ik \cdot s} \log \frac{1}{\xi} + c_0, \quad (6.3)$$

$$c_0 = a_{\text{reg}} - (1 - e^{ik \cdot s}) \ell. \quad (6.4)$$

Meaning. At $\xi = 0$ we have $M = 0$, and $\frac{1}{D} = \frac{1}{(r-s)^2}$ is log-singular at $r = s$. For small ξ this gives $e^{ik \cdot s} \int d^2 u \frac{1}{u^2 + M^2} \sim \pi e^{ik \cdot s} \log \frac{1}{\xi}$. In the test case $e^{ik \cdot s} = 0.75836 + 0.65183i$ and $c_0 = -0.66343 - 0.24594i$:

ξ	$G(\xi)$	$e^{ik \cdot s} \log \frac{1}{\xi} + c_0$	difference
10^{-2}	$2.85266 + 2.76828i$	$2.82896 + 2.75587i$	$2.7 \cdot 10^{-2}$
10^{-4}	$6.32169 + 5.75789i$	$6.32134 + 5.75767i$	$4.1 \cdot 10^{-4}$
10^{-6}	$9.813731 + 8.759483i$	$9.813727 + 8.759480i$	$5.5 \cdot 10^{-6}$

The differences fall like $\xi \log \frac{1}{\xi}$. After multiplying by $C_{UV} \sim \frac{1}{\xi}$ they are integrable.

7. p^+ integral of the other terms: the plus prescription

Both ends carry a $\frac{\log}{\text{pole}}$, so each end needs two distributions. Subtract the endpoint behaviour and add it back:

$$C_{UV}G = \underbrace{\left[C_{UV}G - \frac{e^{ik \cdot s} \log \frac{1}{\xi} + c_0}{\xi} - \frac{\log \frac{1}{\xi}}{\bar{\xi}} \right]}_{\text{integrable on } [0,1]} + e^{ik \cdot s} \frac{\log \frac{1}{\xi}}{\xi} + \frac{c_0}{\xi} + \frac{\log \frac{1}{\xi}}{\bar{\xi}}.$$

(The $O(1)$ part of $C_{UV} = \frac{1}{\xi} - 1 + \dots$ times $\log \frac{1}{\xi}$ is integrable, so it stays inside the bracket.) The elementary integrals are

$$\int_{\lambda}^{1-\lambda} \frac{d\xi}{\xi} = \log \frac{1-\lambda}{\lambda} \rightarrow L, \quad \int_{\lambda}^{1-\lambda} \frac{\log \frac{1}{\xi}}{\xi} d\xi = \left[-\frac{1}{2} \log^2 \xi \right]_{\lambda}^{1-\lambda} \rightarrow \frac{1}{2} L^2,$$

and the same for $\bar{\xi}$. In distribution language:

$$\frac{1}{\xi} = \left[\frac{1}{\xi} \right]_+ + \delta(\xi) L, \quad \frac{\log \frac{1}{\xi}}{\xi} = \left[\frac{\log \frac{1}{\xi}}{\xi} \right]_+ + \frac{1}{2} \delta(\xi) L^2, \quad \frac{\log \frac{1}{\bar{\xi}}}{\bar{\xi}} = \left[\frac{\log \frac{1}{\bar{\xi}}}{\bar{\xi}} \right]_+ + \frac{1}{2} \delta(\bar{\xi}) L^2.$$

$$\begin{aligned} \int_{\lambda}^{1-\lambda} d\xi C_{\text{UV}}(\xi) G(\xi) &= \frac{1+e^{ik \cdot s}}{2} L^2 + c_0 L + H + O(\lambda \log^2 \lambda), \\ H &= \int_0^1 d\xi \left[C_{\text{UV}}(\xi) G(\xi) - \frac{e^{ik \cdot s} \log \frac{1}{\xi} + c_0}{\xi} - \frac{\log \frac{1}{\xi}}{\xi} \right]. \end{aligned} \quad (7.3)$$

Check. In the test case $H = 0.12277 + 0.36849i$. Direct integration of $C_{\text{UV}}G$ over $[\lambda, 1-\lambda]$ compared with the formula:

λ	$\int C_{\text{UV}}G$ (direct)	$\frac{1+e^{ik \cdot s}}{2} L^2 + c_0 L + H$	difference
10^{-3}	$37.50412 + 14.22366i$	$37.49192 + 14.22142i$	$1.2 \cdot 10^{-2}$
10^{-4}	$68.59525 + 25.75127i$	$68.59361 + 25.75098i$	$1.7 \cdot 10^{-3}$
10^{-5}	$109.01816 + 40.73650i$	$109.01795 + 40.73649i$	$2.1 \cdot 10^{-4}$

8. Final answer

Put §5 and §7 into §4:

$$\begin{aligned} \frac{d^3 N_{2b}^{\text{UV}}}{d^3 k} &= \frac{1}{(2\pi)^3} \frac{g^4}{16\pi^4} \frac{1}{k^+} \int_{x,y,x',w'} e^{-ik \cdot w'} e^{ik \cdot x} \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2} \left\{ \left(2L - \frac{11}{6} \right) \left[\frac{1}{\epsilon_{\text{UV}}} + \log \frac{\mu_{\text{MS}}^2 (y - x)^2 e^{2\gamma_E}}{4} \right] \right. \\ &\quad \left. + \frac{1+e^{ik \cdot (y-x)}}{2} L^2 + c_0 L + H \right\} \left[U^{ab'}(x') - U^{ab'}(w') \right] \left[-N_c U^{af}(x) + N_c U^{af}(y) \right] \rho^{b'}(x') \rho^f(y) \end{aligned}$$

- **UV term:** $\left(2L - \frac{11}{6} \right) \left[\frac{1}{\epsilon_{\text{UV}}} + \log \frac{\mu_{\text{MS}}^2 (y-x)^2 e^{2\gamma_E}}{4} \right]$. The coefficient is $\int_{\lambda}^{1-\lambda} C_{\text{UV}} = \int \frac{P_{gg}}{2N_c}$.
- **Other terms:** $\frac{1+e^{ik \cdot (y-x)}}{2} L^2 + c_0 L + H$. Here c_0 is (6.4) and H is (7.3); both are finite and free of ϵ and Λ .
- **Momentum form**, if preferred: replace the log by $\log \frac{\mu_{\text{MS}}^2}{k^2}$ and add $\left(2L - \frac{11}{6} \right) \ell$ to the other terms.

9. Important: the $\frac{1}{2} e^{ik \cdot (y-x)} L^2$ cancels against the rest of 2b

This double log comes from the region $r \rightarrow s$ at $\xi \rightarrow 0$ (eq. 6.3). It is **not** in the full 2b:

- The only block of 2b enhanced by $\frac{1}{\xi}$ is $T_4 = \frac{\bar{\xi}}{\xi} \frac{P \cdot (s-r)}{P^2 r^2} \left[1 - \frac{s \cdot r}{2s^2} \right]$. It **vanishes at** $r = s$ (sympy: $T_4|_{r=s} = 0$), so the full 2b has no $\log \frac{1}{M^2}$ there.
- Your piece has $\frac{\bar{\xi}}{\xi} \frac{P \cdot s}{P^2 r^2}$ instead, which is $\frac{\bar{\xi}}{\xi} \mathcal{W} \neq 0$ at $r = s$.
- So the rest of 2b (full – your piece) carries $-\mathcal{W} e^{ik \cdot s} \frac{\log(1/\xi)}{\xi}$, which gives $-\frac{1}{2} e^{ik \cdot s} \mathcal{W} L^2$. This agrees with the earlier finding that the full 2b remainder has only a simple pole at $\xi \rightarrow 0$ ($\xi \mathcal{I}_b \rightarrow \beta_0$: 2.60199, 2.61189, 2.61435, 2.61517, 2.61539).

Bookkeeping of the double logs in 2b:

	$\xi \rightarrow 0$	$\xi \rightarrow 1$
this piece	$+\frac{1}{2} e^{ik \cdot s} \mathcal{W} L^2$	$+\frac{1}{2} \mathcal{W} L^2$
rest of 2b	$-\frac{1}{2} e^{ik \cdot s} \mathcal{W} L^2$	0
full 2b	0	$\frac{1}{2} \mathcal{W} L^2$

The full-2b column agrees with `rowV_full_derivation.pdf` and `rowV_coordinate_UV.pdf`. The $\xi \rightarrow 1$ entry of the rest follows from those results: T_2 's bracket $\rightarrow \frac{1}{2}$ at large r removes half, and T_3, T_4, T_6 between M and $\frac{1}{q}$ put it back.

Practical advice. Either:

- keep this split, but **do not** plus-prescribe the $e^{ik \cdot s \frac{\log(1/\xi)}{\xi}}$ separately in each half — add the two halves first; or
- subtract a UV term without $\frac{1}{D}$, such as $\frac{C_{UV} W s^2}{r^2(r^2+s^2)}$ (`rowV_coordinate_UV.pdf`). It has no $r = s$ singularity, so no artefact appears.

Either way, the UV term, $\mathcal{W}(2L - \frac{11}{6}) [\frac{1}{\epsilon_{UV}} + \log \frac{\mu^2(y-x)^2 e^{2\gamma_E}}{4}]$, is the same.

Checks: `src/rowV_uvpiece.py`, output in `RESULTS_rowV_uvpiece.txt`. It covers the prefactor and the identity 3.1 (sympy), the closed form against direct 2D quadrature, both endpoint limits, the ξ integral with the plus prescription at three values of λ , and $T_4|_{r=s} = 0$.